



Fitness Planner

CHINONSO OBASI

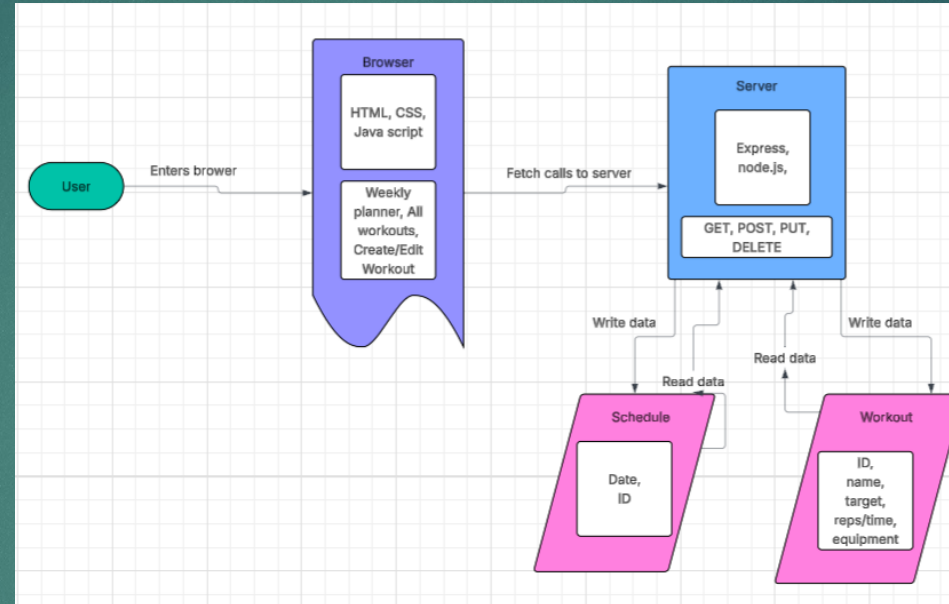
G00403720

BENG (HONS) SOFTWARE & ELECTRONIC ENGINEERING

Summary

This project is a web-based workout planner that allows users to create, manage, and schedule workouts throughout the week. Users can add exercises with details like target area, reps, and equipment, then assign them to specific days using an interactive weekly planner. The app uses a Node.js and Express backend to store data in JSON files, ensuring workouts and schedules are saved.

Architecture diagram



Results

FITNESS PLANNER
workouts created: 4

Weekly Planner

Schedule workouts for each day of the week.

←

→

Create Workout

All Workouts

Monday 27 Apr	Tuesday 28 Apr	Wednesday 29 Apr	Thursday 30 Apr	Friday 1 May	Saturday 2 May	Sunday 3 May
sit ups 3x10 ×	push ups 3x12 ×	+ Add	+ Add	+ Add	+ Add	+ Add
bicep curls 3x10 ×	sit ups 3x10 ×					
+ Add	plank 1min ×					
	+ Add					

Technologies

Software:

Node.js: Runtime environment that runs your backend server.

Express:

Framework used to build the server and handle API routes.

Tools:

HTML: Used to structure the layout of the app.

CSS: Used to style and design the interface.

JavaScript:

Handles frontend logic and interactivity.

Fetch API: Allows communication between frontend and backend.

Json: Stores

Project Management



Attended a few stand-ups where I would share how much progress I made working on my project.



Communicated a bit with my supervisor showing them what I had been working on and also asked questions and received feedback on work.



Kept track of progress using jira and github.



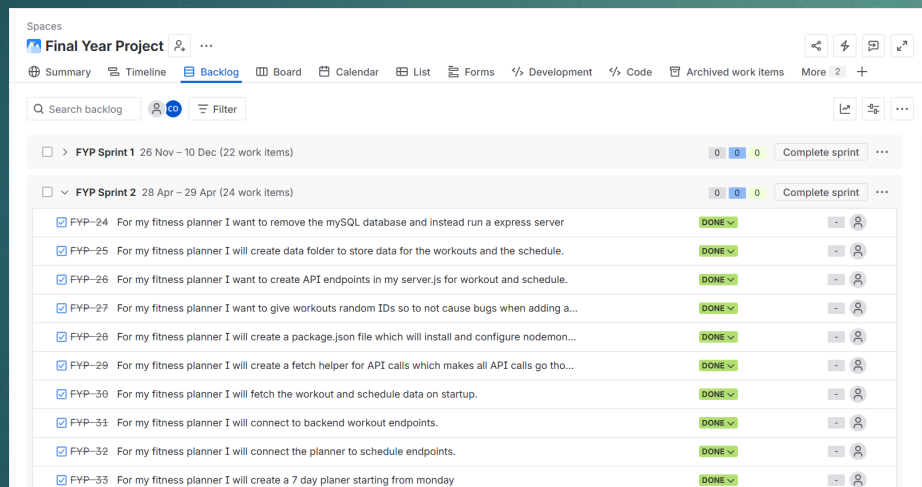
On jira I made sprints which would keep track of all the changes I made to the fitness planner.



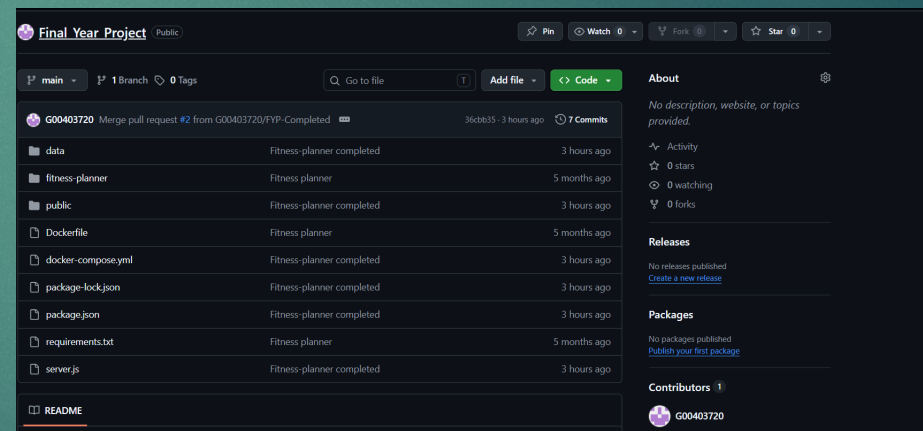
I would push the changes to github with a commit of what was done.

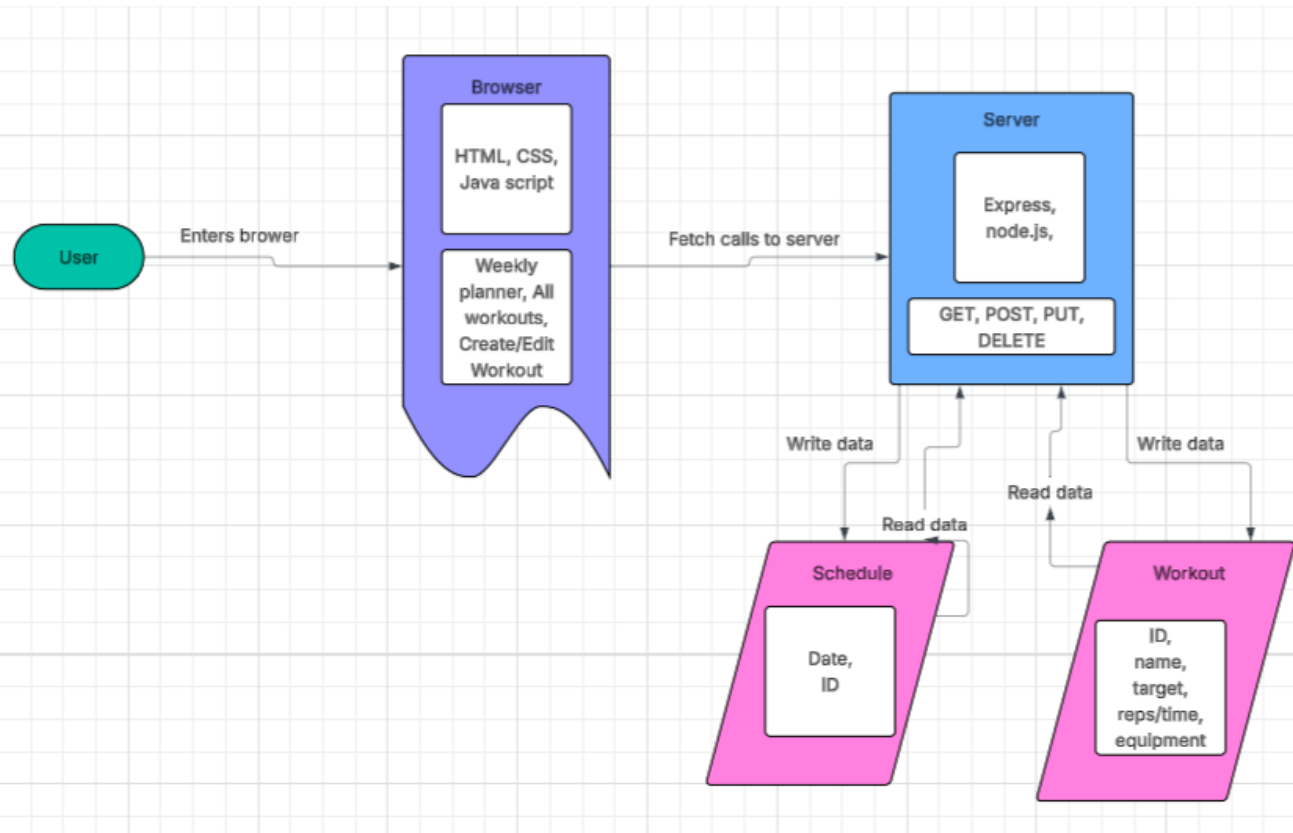
Project Management

Jira



Github





Architecture Diagram

Research & Problem Solving

- ▶ I already had some experience using node.js and express so I didn't have much trouble when setting up the server and web page.
- ▶ Some features like the weekly planner, week navigation and search bar I would ask AI how to create it and what it does.
- ▶ Originally I planned to use a SQL database to save the workout planner but I had trouble with the API endpoints not working so instead I removed the database and created 2 JSON files which would save the workout details and the day a workout



```
data > {} workouts.json > ...
```

```
1  [
2  {
3    "id": "mo35j1waelcaw",
4    "name": "push ups",
5    "target": "Chest",
6    "reps": "3x12",
7    "equipment": "",
8    "notes": ""
9  },
10 {
11   "id": "mo3j9tqmig5rx",
12   "name": "sit ups",
13   "target": "Core / Abs",
14   "reps": "3x10",
15   "equipment": "none",
16   "notes": ""
17 },
18 {
19   "id": "mod0cthp00lvv",
20   "name": "plank",
21   "target": "Core / Abs",
22   "reps": "1min",
23   "equipment": "",
24   "notes": ""

```

```
data > {} schedule.json > ...
```

```
1  {
2    "2026-04-12": [],
3    "2026-04-13": [],
4    "2026-04-14": [],
5    "2026-04-16": [],
6    "2026-04-17": [],
7    "2026-04-20": [],
8    "2026-04-21": [],
9    "2026-04-26": [
10     "mo3j9tqmig5rx",
11     "mod0kyi1y3rjv"
12   ],
13   "2026-05-02": [],
14   "2026-04-27": [
15     "mo35j1waelcaw",
16     "mo3j9tqmig5rx",
17     "mod0cthp00lvv"
18   ]
19 }
```